Technical Report

Multi-Clock-Domain Streaming System

Target Device: Altera Cyclone IV E — EP4CE115F29C7
Tool: Quartus Prime Lite 25.1std

1 Resource Utilization

Synthesis and place-and-route completed with no errors. Table 1 summarises logic resource usage per module.

Table 1: Resource Utilization by Module

Module	Logic Elements	Registers (FFs)	Memory Bits	DSP Blocks
`fir`	58	46	0	0
`afifo`	164	128	0	0
`uart`	50	34	0	0
`power_ctrl`	7	5	0	0
Total (my_sys)	**277**	**213**	**0**	**0**
Device capacity	114,480	114,480	3,981,312	—
Utilization	**<1%**	**<1%**	**0%**	**0%**

The async FIFO dominates resource usage at 164 LEs and 128 registers, consistent with its register-based implementation: all FIFO storage resides in flip-flops rather than block RAM (M9K), so memory bits are zero by design. The FIR filter uses no DSP blocks, confirming that the four multiply-accumulate operations were absorbed into LUT logic as intended. No embedded multipliers or memory blocks were inferred by any module.

2 Static Timing Analysis

2.1 Clock Definitions and CDC Constraints

Two clocks were defined in `my_sys.sdc`: `fast_clk` at 100 MHz (10 ns period) and `slow_clk` at 10 MHz (100 ns period). The domains were declared asynchronous via `set_clock_groups -asynchronous`, automatically cutting all cross-domain paths from STA. The first-stage synchroniser registers at the FIFO boundary were additionally false-pathed with `set_false_path -to`, targeting `*afifo*wptr_sync1_r*` and `*afifo*rptr_sync1_w*`. The Unconstrained Paths report confirmed zero unconstrained endpoints after SDC application, validating full clock coverage.

2.2 Timing Closure Results

Timing closure was achieved on both domains under the worst-case corner (Slow 1200 mV / 85 ° C model), as shown in Table 2.

The critical path lies in the `fast_clk` domain, limited by combinational logic depth through the FIR datapath. A slack of +3.77 ns represents a 37% margin on the 10 ns budget. The `slow_clk`

Table 2: Worst-Case Setup Slack by Clock Domain

Clock	Period	Worst-Case Setup Slack	Status
`fast_clk`	10.00 ns	+3.77 ns	Met
`slow_clk`	100.00 ns	+89.37 ns	Met

domain shows near-full margin, as the UART FSM and baud-rate counter present minimal logic depth relative to the 100 ns budget.

3 Power Analysis — Clock Gating Evaluation

The Power Analyzer was run twice using vectorless estimation to evaluate the effect of the `power_ctrl` clock-gating block and the `ALTCLKCTRL` IP instance. Run 1 applied the default 12.5% toggle rate to all signals, representing the design under active streaming. Run 2 reduced the toggle rate to 1%, approximating the idle state in which `power_ctrl` has de-asserted `clk_en` and `ALTCLKCTRL` has stopped the gated clock. Results are summarised in Table 3.

Table 3: Power Analysis: Active vs. Clock-Gated Idle

Power Component	Run 1 (Active)	Run 2 (Gated)	Delta
Core Dynamic	9.80 mW	9.65 mW	−0.15 mW
Core Static	98.51 mW	98.51 mW	0
I/O	34.01 mW	34.01 mW	0
Total	**142.32 mW**	**142.17 mW**	**−0.15 mW**

Clock gating reduced core dynamic power by 0.15 mW (1.5%). The absolute saving is small because static leakage (98.51 mW) from unused FPGA fabric completely dominates total power consumption. The EP4CE115 contains 114,480 LEs of which fewer than 1% are active; all unused cells contribute leakage current regardless of clock activity. I/O power (34.01 mW) reflects undriven pin termination and is unaffected by internal gating.

This result is architecturally correct. Clock gating eliminates switching activity in the fast-clock domain during idle periods, but on an FPGA the absolute saving is proportional only to the active fabric. The same technique on an ASIC implementation — where static power scales with active cell count rather than total device capacity — would yield proportionally larger savings, potentially reducing dynamic power by the full ratio between active and idle toggle rates.

4 Appendix — Quartus Report Screenshots

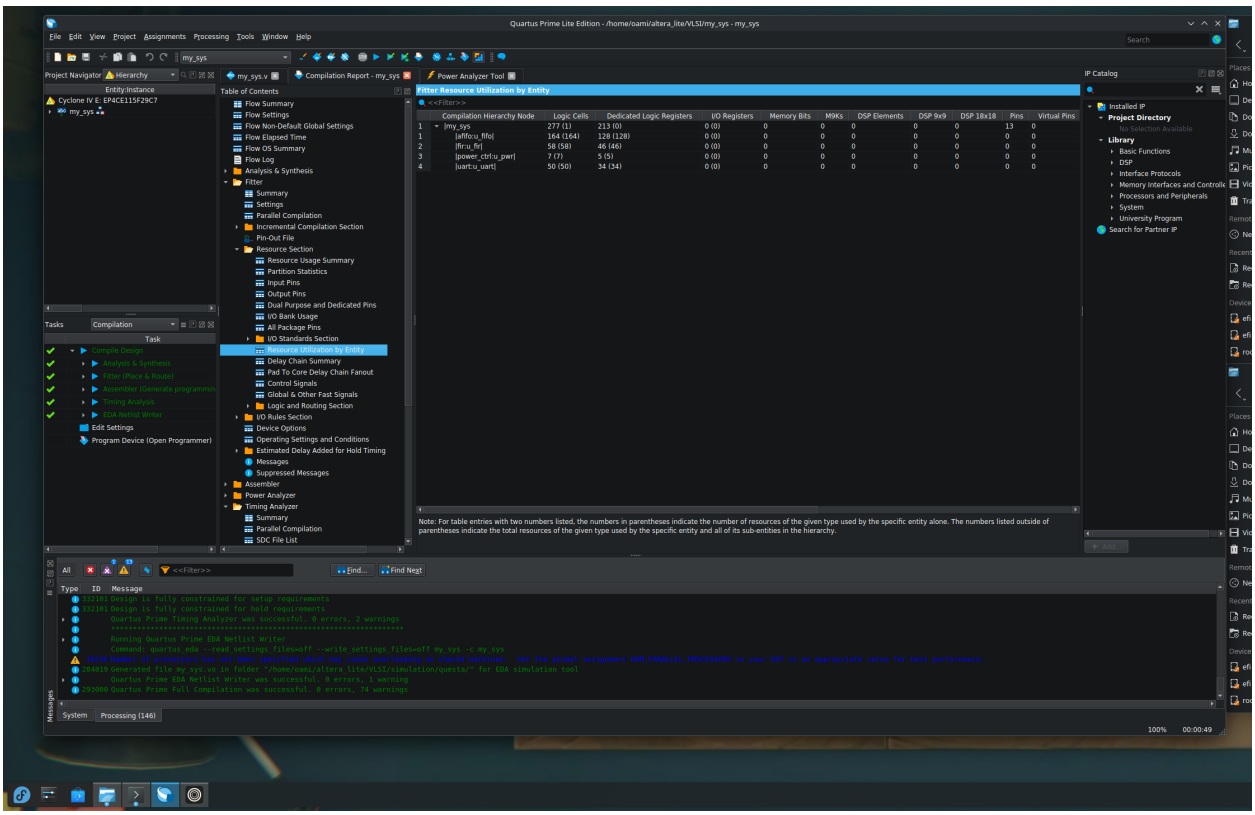


Figure 1: Fitter — Resource Utilization by Entity

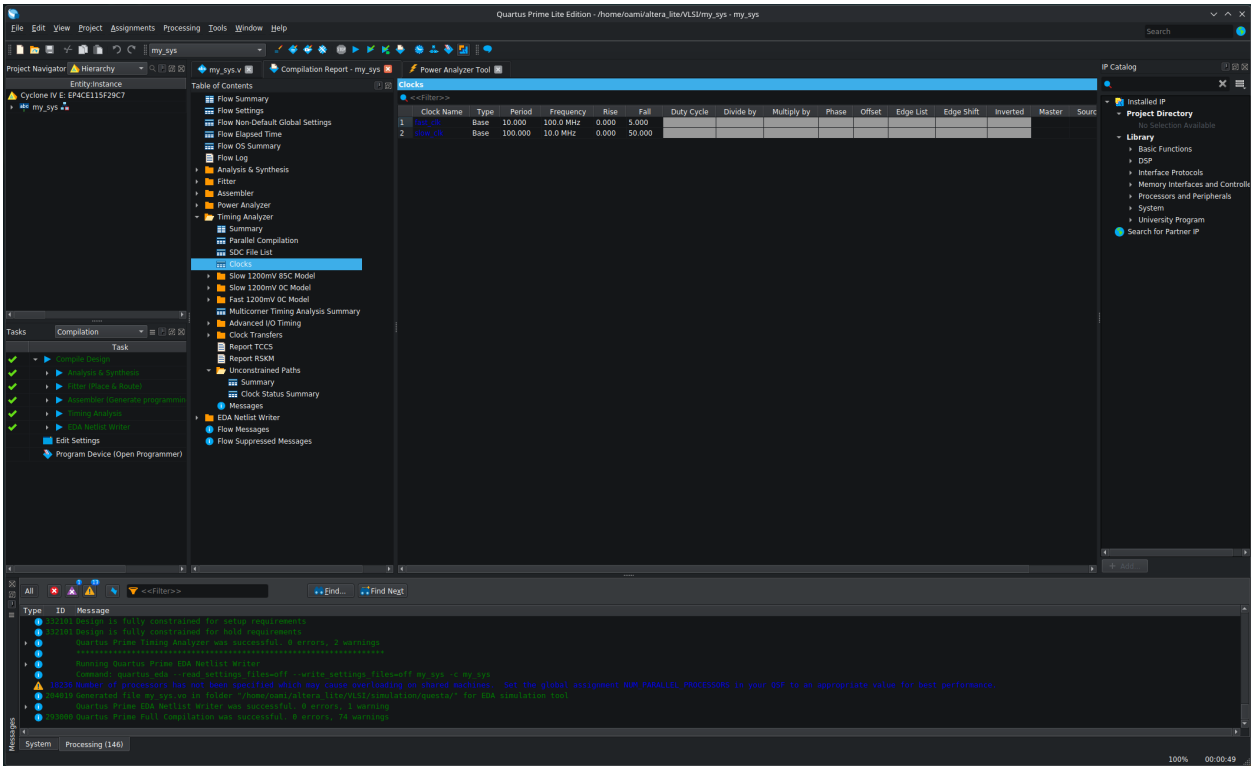


Figure 2: Timing Analyzer — Clock Definitions and Setup Summary

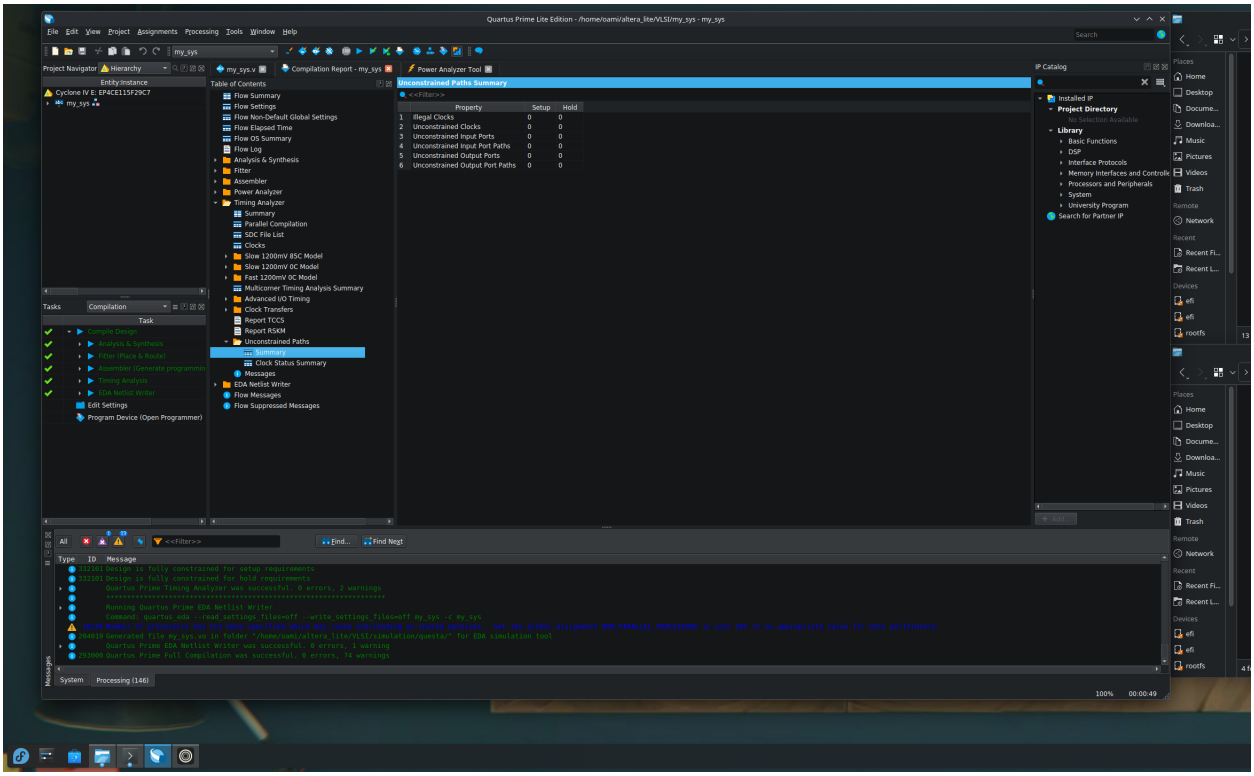


Figure 3: Timing Analyzer — Unconstrained Paths Summary (CDC paths cut)

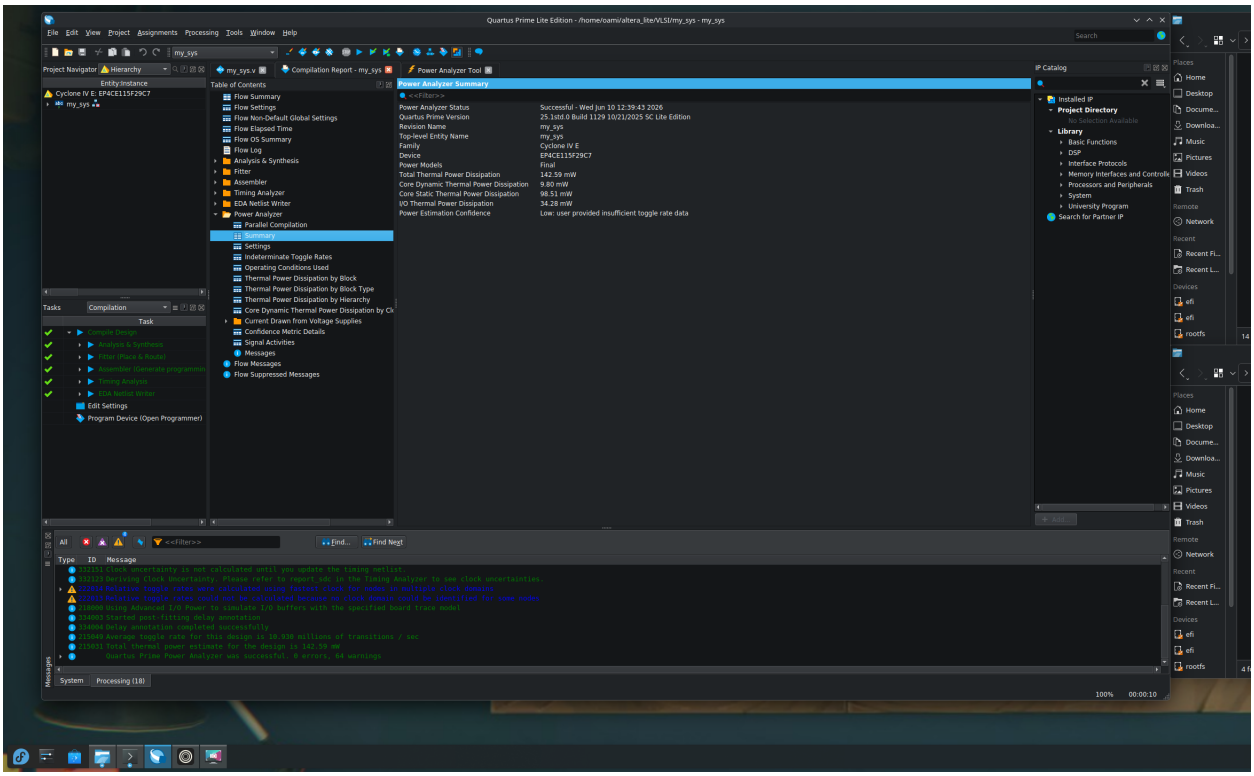


Figure 4: Power Analyzer — Run 1 (Active, 12.5% toggle rate)

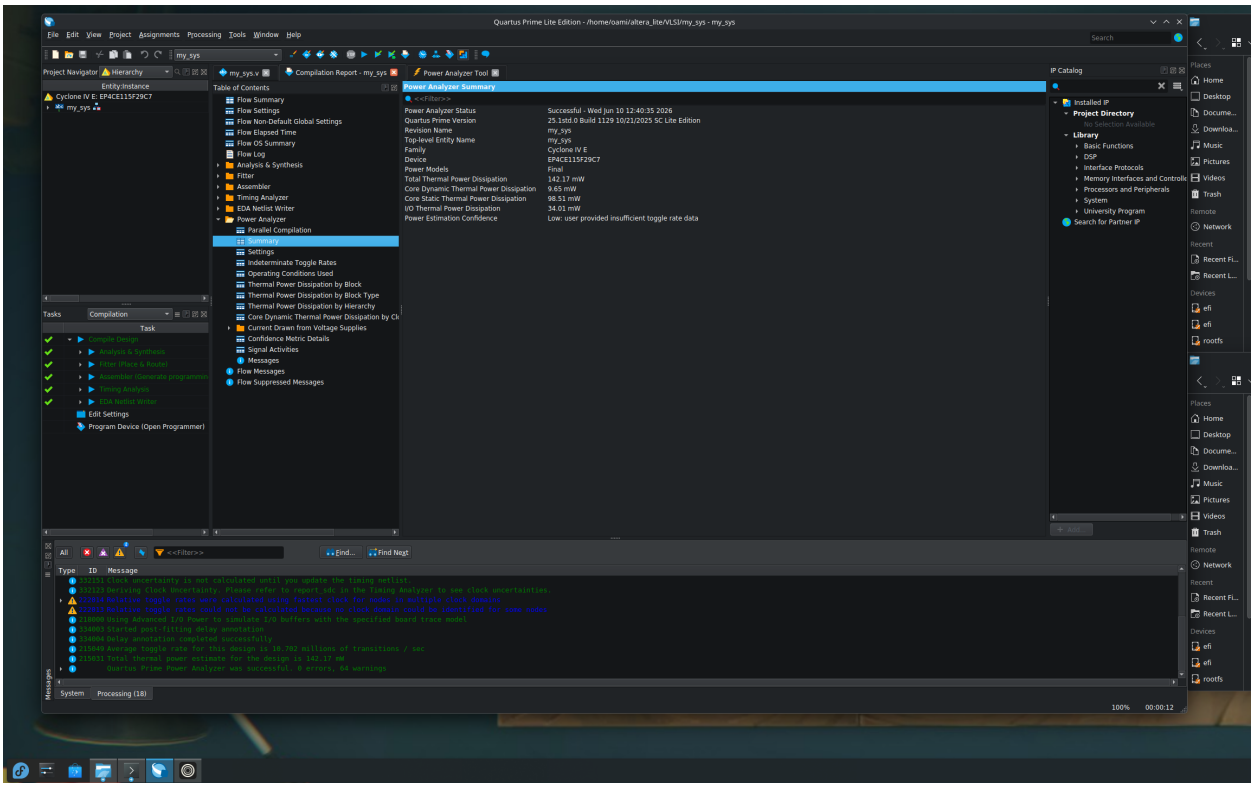


Figure 5: Power Analyzer — Run 2 (Gated/Idle, 1% toggle rate)